AN

INTERNSHIP REPORT

On

AIoT driven Smart Gardening System

Submitted

by

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(Enrollment Number: 221430142036)

Guided By:

Prof. Roshni Mandli

An

Internship Report

Submitted to

Gujarat Technological University

In fulfillment for the award of degree of Bachelor of Engineering

in

CSE(AIML)

ACADEMIC YEAR – 2026

NEW L. J. INSTITUTE OF ENGINEERING AND TECHNOLOGY

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NEW L. J. INSTITUTE OF ENGINEERING AND TECHNOLOGY

CERTIFICATE

This is to certify that the Internship report submitted along with the project entitled Internship in **GTU-School of Engineering and Technology** has been Completed by **RATHOD NIRMIT MUKESHKUMAR** under my guidance in complete fulfilment for the Bachelor of Engineering in **CSE(AIML)** Branch, 8th Semester of **Gujarat Technological University**, Ahmedabad during the academic year 2026.

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Seal of Institute

GUJARAT TECHNOLOGICAL UNIVERSITY

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NAAC Accredited A Grade

Ref. No. : GTU/SET/2026/04/13/AI/04

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for successfully completed **12-week Internship** on

"Foundations to Frontiers AI Internship 2026 "

organized by **GTU - School of Engineering and Technology (Centre of Excellence in AI)**

(Duration 19th January to 19th April , 2026)

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NEW L. J. INSTITUTE OF ENGINEERING AND TECHNOLOGY

DECLARATION

We hereby declare that the Internship report submitted along with the Internship entitled Project **AIoT driven Smart Gardening System** submitted in Complete for Bachelor of Engineering in **CSE(AIML)** Branch to **Gujarat Technological University**, Ahmedabad, is a bonafide record of original Internship work Completed by me at **GTU-School of Engineering and Technology** under the supervision of **External Guide Dr. Seema Joshi** and that no part of this report has been directly copied from any students' reports or taken from any other source, without providing due reference.

Name of Student **Signature of Student**

RATHOD NIRMIT MUKESHKUMAR

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I wish to express my sincere gratitude to my **External guide Dr. Seema Joshi** for continuously guiding me at the company and answering all my doubts with patience. I would also like to thank **Dr. Gayatri Pandi (H.O.D. of CSE, CSE(AIML) Department)** for motivating me every time whenever I get confused, I would also like to thank my **Internal Guide Prof. Roshni Mandli** for helping me through my internship by giving me the necessary suggestions and advices along with their valuable co-ordination in completing this Internship.

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Thus, in conclusion to the above said, I once again thank the staff members of **GTU-School of Engineering and Technology** for their valuable support in completion of the Internship.

Thank You

Name of Student **Date:**

RATHOD NIRMIT MUKESHKUMAR

Enrollment Number **Signature of Student**

221430142036

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AIoT driven Smart Gardening System

Enrollment No.: 221430142036

Student Name: RATHOD NIRMIT MUKESHKUMAR

NEW L. J. INSTITUTE OF ENGINEERING AND TECHNOLOGY (College Code:143)

Semester: VIII, CSE(AIML) Department

ABSTRACT

We are developing a AIoT driven Smart Gardening System for the Company. AIoT driven Smart Gardening System is an immense combination of the Internet of Things (IoT) and Artificial Intelligence (AI) which delivers personalized plant care along with irrigation related solutions. The primary aim for AIoT driven Smart Gardening System is to monitor home gardening activities carried out for efficient plant cultivation. This system is integrated with distributed environmental sensors for continuous capture key parameters like soil moisture, temperature, humidity, while Random Forest Classifier and Regressor model analyze this real-time data for predictive weather aware irrigation, intelligent plant health assessment, and anomaly detection. While live charts displaying sensor readings history and ML assessment log were displayed to monitor and detect any anomalies before happening. System Alerts are constantly delivered to check ongoing activities. This AIoT-enabled smart gardening framework not only enhance plant growth and sustainability but also gives modern, personalized agriculture practices that improve efficiency while minimizing human effort.

Keywords: AIoT (Artificial Intelligence of Things), Smart Gardening System, Environmental Sensors, Random Forest Classifier, Random Forest Regressor, Weather-aware Irrigation, Plant Health Assessment, Sensor Readings, Automation, Anomaly, ML assessment log.

CHAPTER: 1

INTRODUCTION OF PROJECT & COMPANY PROFILE

1.1 Introduction

1.1.1 Overview of the Company

1.1.2 History

1.1.3 Different product / scope of work

1.1.4 Organization Chart

1.1.5 Capacity of Plant

1.1.6 Overview of different plant/unit/department/shop

1.1.7 Details about the work being carried out in each

1.1.8 List the technical specifications

1.1.9 Prepare schematic layout

1.1.10 Explain in details about each stage of production

1.2 Introduction of the Project

1.2.1 Purpose of the Project

1.2.2 Function Requirements

1.2.3 Problems in Existing System

1.2.4 Main Modules

1.1 Introduction

1.1.1 Overview of the Company

Gujarat Technological University (GTU) is a state university established by the Government of Gujarat to oversee and coordinate technical education across the state. The university affiliates a large number of engineering, pharmacy, management, architecture, and other professional colleges in Gujarat.

GTU focuses on:

- Academic excellence in technical and professional education
- Industry–academia collaboration
- Research, innovation, and entrepreneurship
- Skill development and outcome-based education
- Promoting startups and incubation activities.

The university functions as an affiliating body, curriculum designer, examination authority, and research promoter for its associated institutions.

1.1.2 History

- **Established:** 2007
- **Established by:** Government of Gujarat through the GTU Act, 2007
- **Purpose of Establishment:** To centralize and improve the quality of technical education in Gujarat.

  Before GTU was formed, technical colleges were affiliated with different universities. The creation of GTU streamlined curriculum design, examinations, evaluation systems, and academic policies under one dedicated technological university.

  Over the years, GTU has introduced:

- Choice Based Credit System (CBCS)
- Industry-oriented curriculum revisions
- Startup and innovation policies
- Collaboration with industries and international universities
- Centers of Excellence (CoE) in emerging technologies such as Artificial Intelligence, Cyber Security

1.1.3 Different product / scope of work

A. Academic Programs (Products Offered)

GTU offers and regulates the following programs through affiliated colleges:

- Undergraduate Programs (B.E., B.Pharm, BBA, BCA, etc.)
- Postgraduate Programs (M.E., M.Pharm, MBA, MCA, etc.)
- Doctoral Programs (Ph.D.)
- Diploma Programs

B. Academic & Administrative Functions

- Curriculum development and revision
- Conducting examinations and result processing
- Academic audits and quality assurance
- Faculty development programs (FDPs)
- Student training and skill enhancement

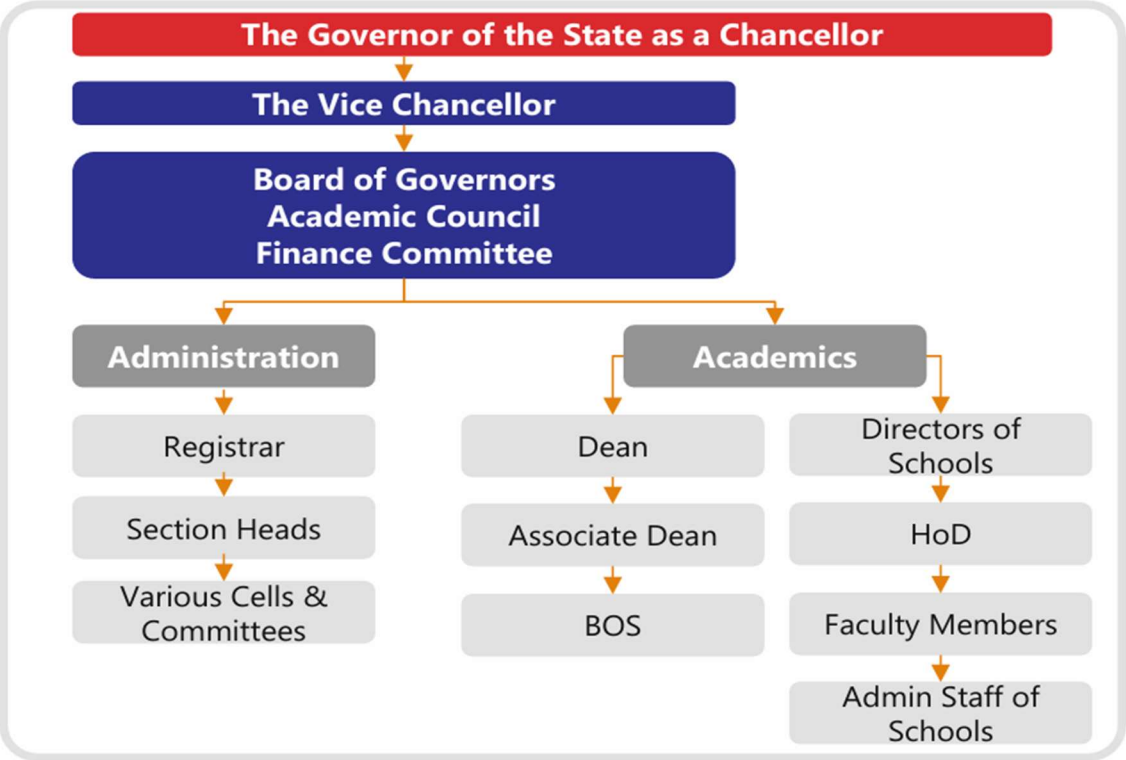
C. Research & Innovation

- Research promotion and Ph.D. supervision
- Startup and incubation support
- Collaboration with industries and research organizations
- Organizing conferences, workshops, and expert talks

D. Industry & Extension Activities

- Industrial visits and internships
- Placement support coordination
- Consultancy and sponsored research projects
- Skill-based certification programs

1.1.4 Organization chart



1.1.5 Capacity of plant

Unlike a manufacturing company, Gujarat Technological University (GTU) is an academic and affiliating university. Therefore, its **"plant capacity"** is defined in terms of:

- Number of affiliated colleges
- Student intake capacity
- Examination handling capacity
- Research and Ph.D. supervision capacity
- Training and skill development output

Approximate Capacity Indicators

- **Affiliated Institutes:** 400+ (Engineering, Pharmacy, Management, etc.)
- **Student Strength:** 3+ lakh students across Gujarat
- **Programs Offered:** Diploma, UG, PG, Ph.D.
- **Examinations Conducted:** Multiple sessions annually (Winter & Summer)
- **Research Capacity:** Hundreds of registered Ph.D. scholars

Thus, GTU's "production capacity" refers to **qualified graduates, research outputs, patents, startups, and skilled professionals**.

1.1.6 Overview of different plant/unit/department/shop of the organization and Layout of the production/process being carried out in company

Since GTU is an educational organization, its departments function like operational units.

Major Departments / Units

1. Academic Section
2. Examination Section
3. Research & Development (R&D) Cell
4. IT & Digital Services Department
5. Finance & Accounts
6. Administration Department
7. Innovation Council & Startup Cell
8. Centers of Excellence (CoE – AI, CoE – Cyber Security, etc.)

1.1.7 It includes the details about the work being carried out in each department.

1. Academic Section

- Curriculum design and revision
- Syllabus updates
- Academic calendar planning
- Faculty coordination

2. Examination Section

- Exam scheduling
- Question paper setting
- Online/offline examination management
- Result processing and declaration

3. Research & Development Cell

- Ph.D. admissions and monitoring
- Research project approvals
- Patent support
- Conference organization

4. IT Department

- TU portal maintenance
- Online exam system
- Result processing software
- Student database management

5. Finance Department

- Budget allocation
- Fee management
- Salary processing

- Audit compliance

6. Innovation & Startup Cell

- Incubation support
- Startup mentoring
- Hackathons and innovation challenges

1.1.8 List the technical specifications of major equipment used in each department.

Since GTU is service-oriented, major equipment includes IT infrastructure.

Department	Major Equipment	Technical Specifications
IT Department	Central Servers	High-performance rack servers, 64–256 GB RAM, RAID storage systems
IT Department	Data Storage Systems	NAS/SAN storage, multi-terabyte capacity
Examination Cell	High-speed Scanners	40–80 ppm document scanners
Examination Cell	OMR Scanners	Optical mark recognition system
Academic Section	Desktop Systems	i5/i7 processors, 8–16 GB RAM
CoE (AI/ML)	AI Workstations	GPU-based systems (NVIDIA RTX/Quadro), 64-128 GB RAM
CoE Labs	IoT Kits	Microcontrollers, sensors, edge devices
Conference Hall	Smart Boards & Projectors	Full HD interactive display systems

1.1.9 Prepare schematic layout which shows the sequence of operation for manufacturing of end product.

Admission of Students

↓

Curriculum Design & Approval

↓

Teaching–Learning Process

↓

Continuous Internal Evaluation

↓

Semester Examination

↓

Evaluation & Result Processing

↓

Award of Degree

↓

Placement / Higher Studies / Startup

1.1.10 Explain in details about each stage of production.

1. Admission Stage

Students enroll through centralized admission committees (ACPC/ACPDC).

2. Curriculum Development

Subject experts and Board of Studies design syllabus aligned with industry requirements.

3. Teaching–Learning Process

- Classroom lectures
- Laboratory sessions
- Project-based learning
- Industrial visits

4. Continuous Internal Evaluation (CIE)

- Assignments
- Mid-semester exams
- Practical assessments
- Project reviews

5. Semester Examination

Conducted centrally by GTU with standardized evaluation.

6. Result Processing

- Digital evaluation
- Moderation
- Result declaration through GTU portal

7. Degree Award

After successful completion of credits, degree certificate is awarded during convocation.

8. Final Output

- Skilled engineers, pharmacists, managers
- Research publications
- Patents
- Startups

1.2 Introduction of the Project

The proposed system implements two AI-enabled features: predictive irrigation, plant health classification using Random Forest classifier and regressor. Sensor data is collected and processed through a machine learning pipeline involving data preprocessing, feature extraction, model training, and real-time prediction. The trained models are integrated with the smart gardening system for automated decision-making and Alerts are displayed on dashboard for simpler analysis which are accessed by user with real time data.

1.2.1 Purpose of the Project

Main purpose for creating this system is promoting the concept of 'Home Gardening', which brings a part of nature into our modern lifestyles. While the world is facing serious global environmental issues gardening at home would make great difference in that.

Additionally, it also helps for de-stressing and pleasant environment in surroundings. But due to lack of awareness and knowledge regarding plant growth patterns and measures the one feels it hard job to carry out such practices at home.

So, there comes this system which helps them as a gardener for cultivating or planting different crops, herbs, vegetables or flowers in outdoor or indoor premises of the house irrespective to climate changes. This is a major step in growth of environmental sustainability at very base level starting from individuals home.

1.2.2 Function Requirements

The functional requirements which should be fulfilled in this projects are as follows:

- Real-time sensor data acquisition
- AI-based irrigation prediction & water usage tracking
- Dashboard monitoring and Plant Health prediction
- User control & notifications including anomaly detections
- Climate friendly plant growing suggestions

1.2.3 Problems in Existing System

Currently the Existing System are facing lots of problem to implement such type of projects in real world but this system focuses on majorly three problems which are mostly seen in existing systems. They are:

1) **Watering System**

   In existing system watering is done manually which results in overwatering or underwatering due to this wastage of water is seen frequently. This system will optimize water usage by adapting environmental changes and fixed interval timer-based irrigation reducing time and water loss.

2) **Plant Health Analysis for their Growth and Yield**

   Plant stress plays a vital role for better yield and overall growth of plant. By measuring real time data through sensor this system evaluates plant health which should be looked after by user on regular basis to keep check on plant.

3) **Anomaly Detection**

   System reliability comes as important part along with such technological advancements. So, considering this measure abnormal behaviors of the sensors installed will be detected and alerts will be generated to overcome in real time.

1.2.4 Main Modules

The flow of System modules is as mentioned below:

Data Acquisition Module (IoT Layer)
↓
Data Management Module (Backend Layer)
↓
AI & Machine Learning Module (Intelligence Layer)
↓
User Interface Module (Application Layer)
↓
Sustainability & Analytics Module

CHAPTER: 2

SYSTEM REQUIREMENTS

2.1 Hardware & Software Requirements

2.1.1 Server-Side Requirements

2.1.2 Developer Side Requirements

2.1.3 User Side Requirements

2. Hardware & Software Requirements

2.1 Server-Side Requirements

- **Hardware Requirement**
  - Arduino UNO / Nano
  - Soil Moisture, DHT Sensors, Relay, DC motor.
  - SD Card, Jumper wires
- **Software Requirement**
  - **Front End:** HTML, CSS, JavaScript
  - **Back End:** MySQL 5.0
  - **Server:** Flask
  - **Design Tool:** Pycharm , Arduino IDE
  - **OS:** Window 7 or Any Compatible OS

2.2 Developer-Side Requirements

- **Hardware Requirement**
  - Arduino UNO / Nano
  - Soil Moisture, DHT Sensors, Relay, DC motor.
  - SD Card, Jumper wires
- **Software Requirement**
  - **Front End:** HTML, CSS, JavaScript
  - **Back End:** MySQL 5.0
  - **Server:** Flask Localhost
  - **Design Tool:** Pycharm , Arduino IDE
  - **OS:** Window 7 or Any Compatible OS

2.3 User-Side Requirements

- **Hardware Requirement**
  - Arduino UNO / Nano
  - Soil Moisture, DHT Sensors, Relay, DC motor.
  - SD Card, Jumper wires
- **Software Requirement**
  - **Front End:** HTML, CSS, JavaScript
  - **Back End:** MySQL 5.0
  - **Server:** Flask localhost
  - **Design Tool:** Pycharm , Arduino IDE

CHAPTER: 3

WORK SHEET REPORT

3.1 Work Sheet Report (15 DAYS)

3.1 Work Sheet Report (15 DAYS):

SUGGESTED 15 DAYS WORK SHEET REPORT	
Student Name:	**RATHOD NIRMIT MUKESHKUMAR**
Enrollment No:	**221430142036**
Internship/Project Title	**AIoT driven Smart Gardening System**
Tools and Technologies	**Kaggle, PyCharm, Python Libraries**
Company/ Organization Name	**GTU-School of Engineering and Technology**

Student's Activity Details:

Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
1	19/01/2026 To 23/01/2026	- Python setup - Data manipulation - Linear and Logistic Regression - Implement 15+ Models	Completed	Good
2	22/01/2024 To 27/01/2024	- Loading and Inspecting Data - Cleaning Data, Handling Missing Values, Encoding - Multivariate analysis - Create new features	Completed	Good

SUGGESTED 15 DAYS WORK SHEET REPORT	
Student Name:	**RATHOD NIRMIT MUKESHKUMAR**
Enrollment No:	**221430142036**
Internship/Project Title	**AIoT driven Smart Gardening System**
Tools and Technologies	**ML Algorithms, Python Libraries**
Company/ Organization Name	**GTU-School of Engineering and Technology**

Student's Activity Details:

Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
3	02/02/2026 To 06/02/2026	- Linear Regression - Logistic Regression - Ridge & Lasso Implementation - Project working model	Completed	Good
4	09/02/2026 To 13/02/2026	-K-fold Cross Validation -Random Forest -ROC & Precision curves -Project further processing	Completed	Good

SUGGESTED 15 DAYS WORK SHEET REPORT	
Student Name:	**RATHOD NIRMIT MUKESHKUMAR**
Enrollment No:	**221430142036**
Internship/Project Title	**AIoT driven Smart Gardening System**
Tools and Technologies	**Model Efficiency and Precision, PyCharm**
Company/ Organization Name	**GTU-School of Engineering and Technology**

Student's Activity Details:

Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
5	16/02/2026 To 20/02/2026	- K-means - DBSCAN - PCA - Industry Expert session - Project working	Completed	Good
6	23/02/2026 To 27/02/2026	- Neural Networks - Backend propagation - CNN application - Project implementation	Completed	Good

SUGGESTED 15 DAYS WORK SHEET REPORT	
Student Name:	**RATHOD NIRMIT MUKESHKUMAR**
Enrollment No:	**221430142036**
Internship/Project Title	**AIoT driven Smart Gardening System**
Tools and Technologies	**Sensor connection, Arduino IDE**
Company/ Organization Name	**GTU-School of Engineering and Technology**

Student's Activity Details:

Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
7	02/03/2026 To 06/03/2026	-Sequential data modelling - Transfer Learning -Industry Expert Session -Project real world implement	Completed	Good
8	09/03/2024 To 13/03/2024	-GAN -DCGAN, CycleGAN -Mode collapse, evaluation -Project implementation	Completed	Good

SUGGESTED 15 DAYS WORK SHEET REPORT	
Student Name:	**RATHOD NIRMIT MUKESHKUMAR**
Enrollment No:	**221430142036**
Internship/Project Title	**AIoT driven Smart Gardening System**
Tools and Technologies	**Arduino IoT kit, Flask**
Company/ Organization Name	**GTU-School of Engineering and Technology**

Student's Activity Details:

Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
9	16/03/2026 To 20/03/2026	- NLP -Tokenization, lemmatization, stemming. - INTRO TO BERT, GPT - Industry visit	Completed	Good
10	23/03/2026 To 27/03/2026	- Meta -Learning - Reinforcement Learning - Interview practice - Expert session	Completed	Good

SUGGESTED 15 DAYS WORK SHEET REPORT	
Student Name:	**RATHOD NIRMIT MUKESHKUMAR**
Enrollment No:	**221430142036**
Internship/Project Title	**AIoT driven Smart Gardening System**
Tools and Technologies	**Arduino IoT kit, PyCharm**
Company/ Organization Name	**GTU-School of Engineering and Technology**

Student's Activity Details:

Week Number	**Start Date to End Date**	**Tasks to be assigned**	**Tasks to be completed**	**Remarks**
11	30/03/2026 To 03/04/2026	-Presentation interaction - Industry visit - Project completion	Completed	Good
12	06/04/2026 To 10/04/2026	- Efficient Research Technologies - Project evaluation fully working model	Completed	Good

CHAPTER: 4

AI MACHINE LEARNING MODEL

4.1 AI/ML Model

4.1.1 MODEL ARCHITECTURE

4.1.2 TRAINING DATA STRUCTURE

4.1.3 FEATURE ENGINEERING

4.1.4 TRAINING PROCESS

4.1.5 PERFORMANCE METRICS

4.1 AI/ML Model

4.1.1 MODEL ARCHITECTURE

The system uses an ensemble of three Random Forest models working together:

- **Classifier**- The Random Forest classifier predicts the health category into different groups "Excellent" ,"Good" ,"Fair" ,"Poor" ,"Critical".
- **Regressor**- It predicts the health score ranging from 0–100 integer.
- **Pump Predictor-** Predicts if irrigation needed considering readings taken from sensors and gives output in Boolean (yes/no).

All three are Random Forest algorithms from scikit-learn, trained on your real sensor data (12,180 readings over 12 weeks).

4.1.2 Training Data Structure

The model is trained on 182,700 samples created from your 12,180 real readings:

Base readings: 12,180 (your actual sensor data) + Plant profiles: 15 (Tomatoes, Basil, Rose, etc.) = Training samples: 182,700 rows.

Each sample contains:

- **Raw sensor values** — moisture %, temp °C, humidity %
- **Rolling statistics** — 3-reading avg, 12-reading avg, std dev, min, max
- **Trend features** — change over last 5 readings (is moisture dropping?)
- **Time features** — hour/day encoded as sine/cosine (captures daily irrigation cycle)
- **Plant-specific features** — distance from that plant's ideal range
- **Interaction features** — compound stress (heat × dryness, VPD proxy)
- **Target labels** — health score (0-100), health label, pump needed (0/1)

4.1.3 FEATURE ENGINEERING

The model doesn't just read raw sensor values — it engineers 28 intelligent unique features:

A) Raw Sensor Readings (3 features):

1) "moisture" [soil moisture percentage (0-100)]
2) "temp" [temperature in Celsius]
3) "humidity" # relative humidity percentage

B) Rolling Window Statistics (18 features)- Captures trends over time — critical for health prediction:

Moisture rolling features
"m_mean_3" [average over last 3 readings (short-term trend)]
"m_mean_12" [average over last 12 readings (long-term trend)]
"m_std_12" [standard deviation (variability)]
"m_trend_5" [change over last 5 readings (moisture / 5)]
"m_min_12" [minimum in last 12 readings (drought detection)]
"m_max_12" [maximum in last 12 readings (flood detection)]

Temperature rolling features
"t_mean_3", "t_mean_12", "t_trend_5"

Humidity rolling features
"h_mean_3", "h_mean_12", "h_trend_5"

A single moisture reading of 45% could be:

- Healthy if it's been stable at 45% for hours
- Concerning if it dropped from 70% to 45% in the last hour

The model sees both the value and the trajectory**.**

C) Time-of-Day Features (4 features):
Captures the daily irrigation pattern (pump fires 3× per day):

D) Plant-Strees features (3 features):
Interaction effects — when two stresses combine, damage is worse:

1) "heat_dryness" = temp × (100 - moisture) / 100
2) "vpd_proxy" = temp × (1 - humidity / 100)
3) "moisture_x_hum" = moisture × humidity / 100

4.1.4 TRAINING PROCESS

The training process carried out while making model is as below:

Step 1 — Data Split

Train: 85% (155,295 samples)

Test: 15% (27,405 samples)

Step 2 — Train Three Models

Classifier: predicts health_label (5 classes)

Regressor: predicts health_score (0-100)

Pump Predictor: predicts pump_needed (binary)

Step 3 — Validation

- **Test set accuracy** — never seen these samples during training
- **5-fold cross-validation** — splits data 5 ways, trains 5 times, averages results

4.1.5 PERFORMANCE METRICS

A) Classifier (Health Label Prediction)

Accuracy: 99.34%

Precision/Recall/F1 per class:

	precision	recall	f1-score	support
Critical	0.93	1.00	0.96	75
Excellent	1.00	0.99	1.00	6619
Fair	1.00	0.98	0.99	11481
Good	0.98	0.99	0.99	8324
Poor	0.95	0.98	0.96	906

- **Precision** — when model says "Excellent", it's right 100% of the time
- **Recall** — model finds 99% of actual "Excellent" cases
- **F1-score** — harmonic mean of precision/recall (balanced metric)

B) Regressor (Health Score Prediction)

MAE (Mean Absolute Error): 0.44 points

RMSE: 0.67 points

R^2: 0.9892

- Average prediction error: ±0.44 points on a 0-100 scale
- If true score is 75, model predicts 74.56–75.44
- $R^2 = 0.9892 \rightarrow$ model explains 98.92% of variance (nearly perfect)

C) Pump Predictor

Accuracy: 97.26%

		Predicted	
		No	Yes
Actual	No	[12450]	[128] ← 128 false alarms (unnecessary pump)
	Yes	[622]	[14205] ← 622 missed pumps (should have fired)

- Missing a pump (622 cases) → plant suffers from drought
- False alarm (128 cases) → wastes water but harmless
- The model is tuned to prefer false alarms over missed pumps

CHAPTER: 5

HARDWARE COMPONENTS

5.1 Arduino Uno

5.2 Soil Moisture Sensor

5.3 DHT11 Sensor

5.4 Relay Module

5.5 Water Pump / Motor

5.1 Arduino Uno

Arduino Uno is a microcontroller-based development board based on the ATmega328P. It is widely used in IoT projects for controlling sensors, actuators, and communication between hardware and software systems.

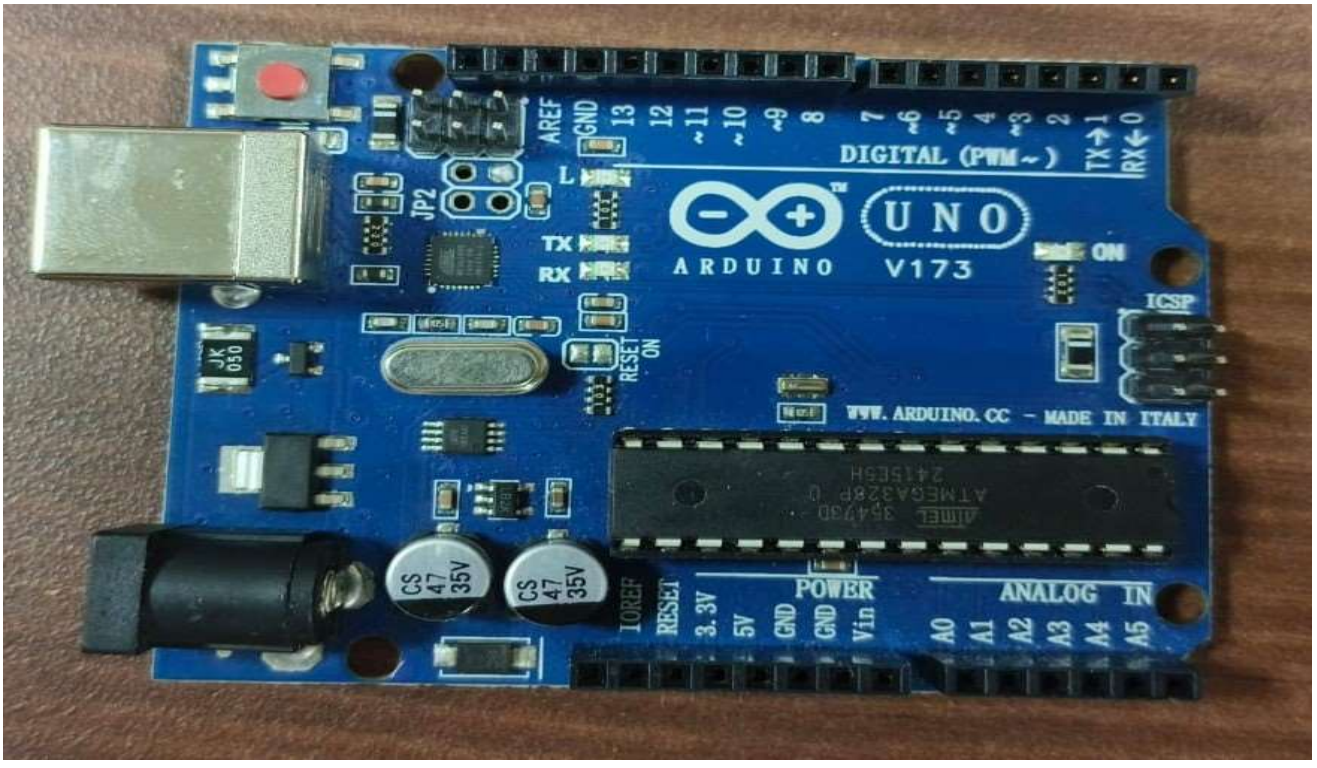
Working Principal:

Arduino Uno reads data from connected sensors (soil moisture, temperature, humidity) through analog and digital pins. It processes this data using ML models and controls output devices such as relay modules for irrigation automation.

Roles of Arduino Uno in Smart Gardening System:

- Collects sensor data (soil moisture, temperature, humidity)
- Sends data to the computer/dashboard (via serial communication)
- Controls water pump using relay
- Acts as the central controller

Arduino Uno image:



5.2 Soil Moisture Sensor

A soil moisture sensor measures the water content in soil, which helps in determining when irrigation is required.

- The Operating Voltage for Soil moisture sensor is 3.3V-5V.
- The Output Type for Soil moisture sensor is in Analog data output.
- The Measurement contains Soil Water contents.

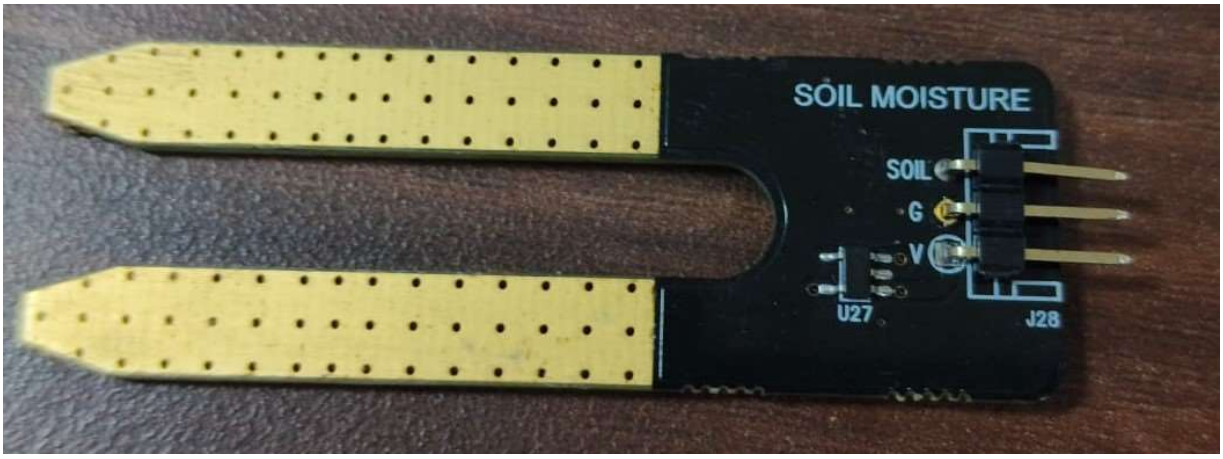
Working Principle

- The sensor has two probes inserted into soil.
- It measures electrical resistance.
- Wet soil → Low resistance → High conductivity.
- Dry soil → High resistance → Low conductivity.

Roles of Soil Moisture Sensor in Project

- Detects soil dryness level
- Triggers irrigation when moisture is below threshold
- Helps automate watering system

Soil moisture sensor image:



5.3 DHT11 Sensor

DHT sensor measures temperature and humidity, which are important environmental parameters for plant growth. It is operated on 3.5V- 5V.

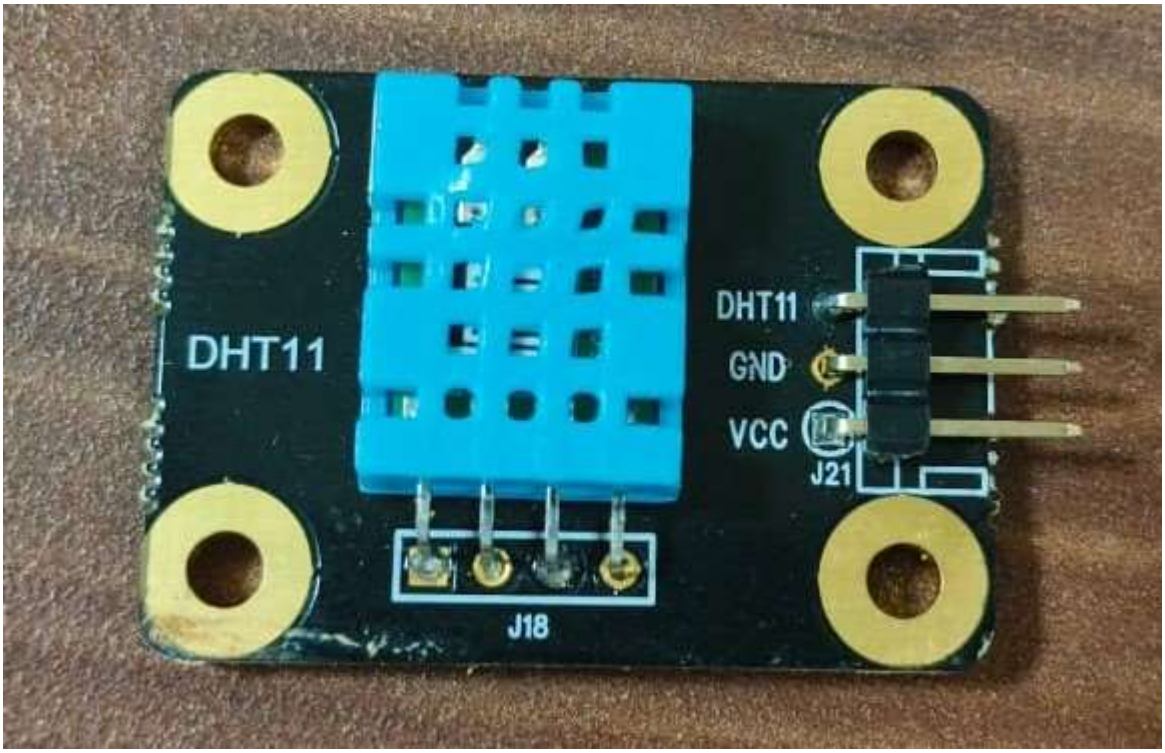
Working Principle

- Uses a capacitive humidity sensor
- Measures moisture in air
- Sends digital signal to Arduino

Roles of DHT11 Sensor in Project

- Monitors environmental conditions
- Helps in plant health analysis
- Data used in AI prediction model

DHT11 Sensor image:



5.4 Relay Module

A relay is an electromechanical switch used to control high voltage devices like a water pump using low voltage signals from Arduino.

Working Principle

- Arduino sends signal to relay
- Relay acts as a switch
- ON → Pump runs
- OFF → Pump stops

Roles of Relay in Project

- Controls irrigation system
- Enables automatic watering
- Provides electrical isolation

5.5 Water Pump / Motor

A water pump is used to supply water to plants during irrigation.

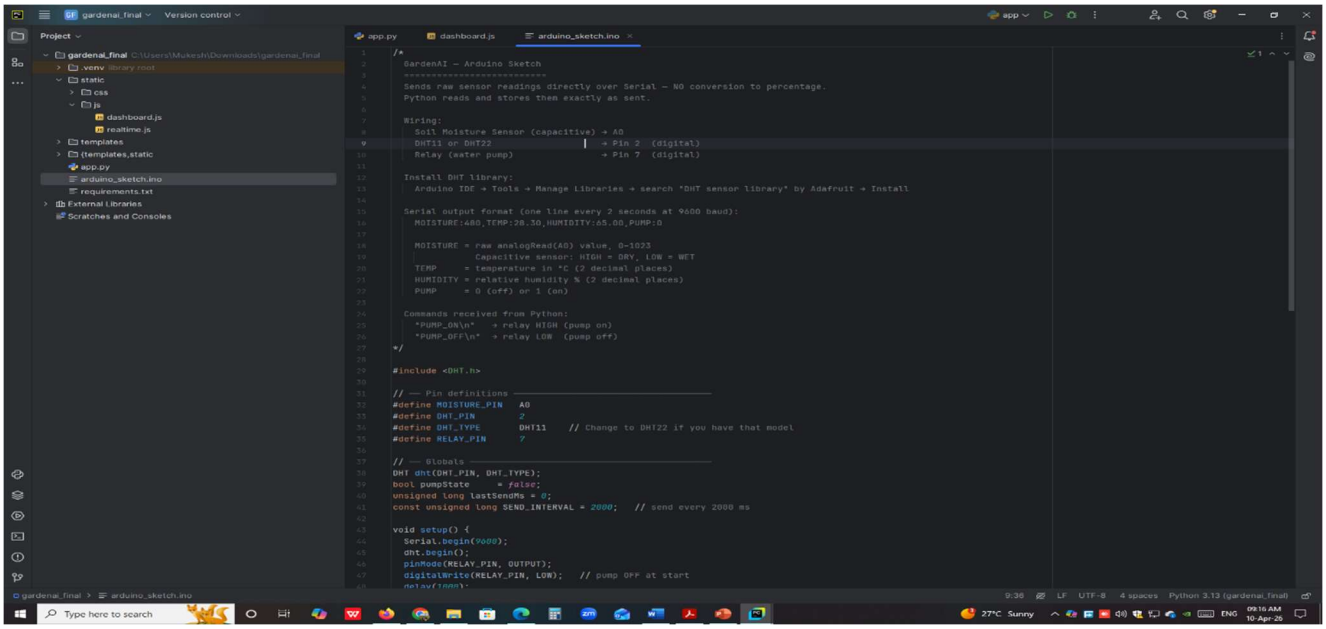
Working Principle

- Activated via relay
- Pumps water from reservoir to plants

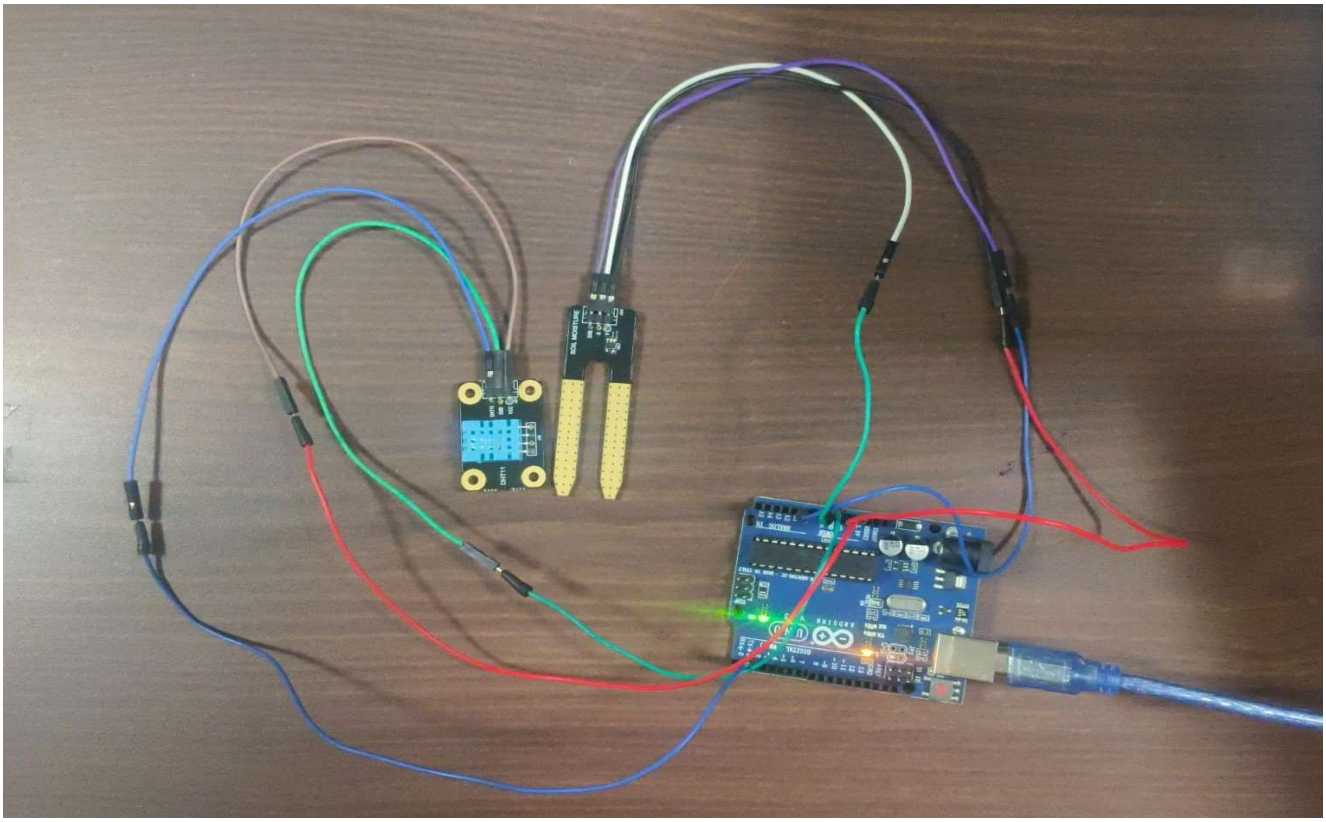
Role in Project

- Executes irrigation process
- Controlled automatically or manually

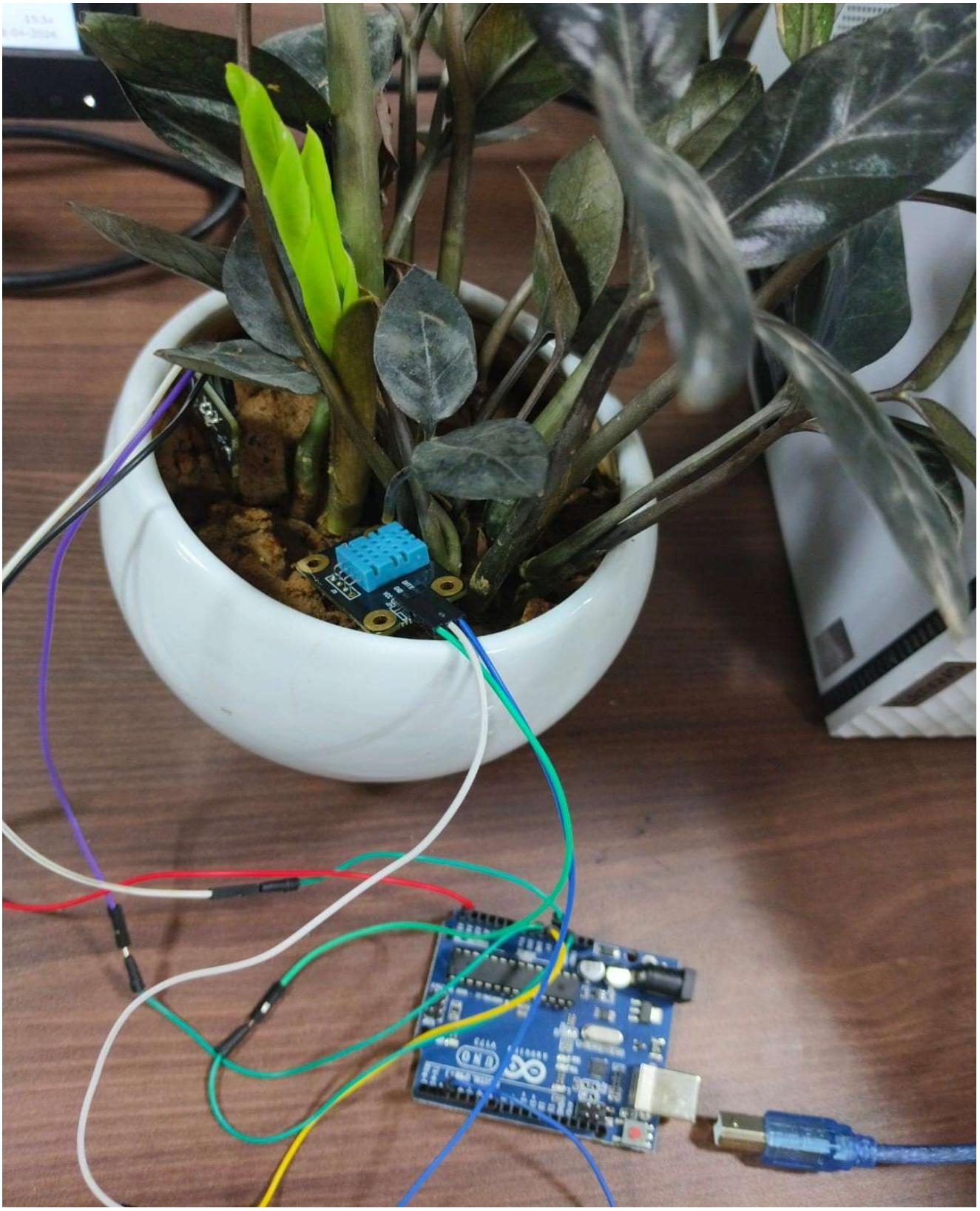
Arduino Sketch:



Whole System image:



Plants Image:



CHAPTER: 6

DATA DICTIONARY

6.1 Introduction

6.2 List of Tables

6.2.1: ai_assessment

6.2.2: sensor_readings

6.2.3: plant_profiles

6.1 Introduction

Data Dictionary is an important part of a project or system which contains all definition of elements in the system. In Data Dictionary you will find a list of all elements composing the data flowing through a system.

- The major elements of a system:
  - Data Flow
  - Data Store
  - Processes
- The data dictionary stores all details and description of these elements.
- The data dictionary provides additional information about the system.
- The data dictionary contains the data about data e.g. Student is a data and this student belongs to this course is a description of STUDENT data which is stored in data dictionary.
- Why is Data Dictionary important?
  - To manage the details in large system.
  - To communicate a common meaning for all system elements.
  - To document the features of system.
  - To determine where system changes should made.
  - To locate errors and omissions in the system.

6.2 List of Tables:

- **Table 6.2.1: ai_assessment**

  Description: This table stores data related to the AI/ML module of the ML assessment results per reading.

[Table 6.2.1: ai_assessment]

Field Name	Data type (size)	Constraint	Description
id	Integer	Primary key, Auto Increment	Stores the unique identifier of the plant.
plant	Varchar(100)	Not Null, Unique	Stores the name of the plant.
moisture	Integer	Not Null	Stores the Moisture.
humidity	Float	Not Null	Store the Humidity.
temperature	Float	Not Null	Stores the Temperature.
healthscore	Integer	Not Null	Stores the predicted health score by ml model.
healthlabel	Varchar(100)	Not Null	Stores the classified health label by ml model.

- **Table 6.2.2: sensor_readings**

  Description: This table stores data related to the history of every Arduino reading

[Table 6.2.2: User_ master]

Field Name	Data type (size)	Constraint	Description
id	Integer	Primary key, Auto Increment	Stores the unique identifier of the user.
moisture	Integer	Not Null	Stores the Moisture.
humidity	Float	Not Null	Store the Humidity.
temperature	Float	Not Null	Stores the Temperature.
recording	Date and Time	Not Null	Store Time Stamps.
pump	Bollean	Not Null	Stores the operation status of pump

- **Table 6.2.3: plant_profiles**

Description: This table stores data related to the user-defined plants with ideal sensor ranges.

[Table 6.2.2: User_ master]

Field Name	Data type (size)	Constraint	Description
id	Integer	Primary key, Auto Increment	Stores the unique identifier of the user.
moisturelow	Integer	Not Null	Stores the lower level of Moisture.
humiditylow	Float	Not Null	Store the lower level of Humidity.
temperaturelow	Float	Not Null	Stores the lower level of Temperature.
moisturehigh	Integer	Not Null	Stores the higher level of Moisture.
humidityhigh	Float	Not Null	Store the higher level of Humidity.
temperaturehigh	Float	Not Null	Stores the higher level of Temperature.
carenotes	Varchar(100)	Not Null	Store the tips for careing plants.

CHAPTER: 7

SNAPSHOT OF WEBSITE

7.1 Home Page

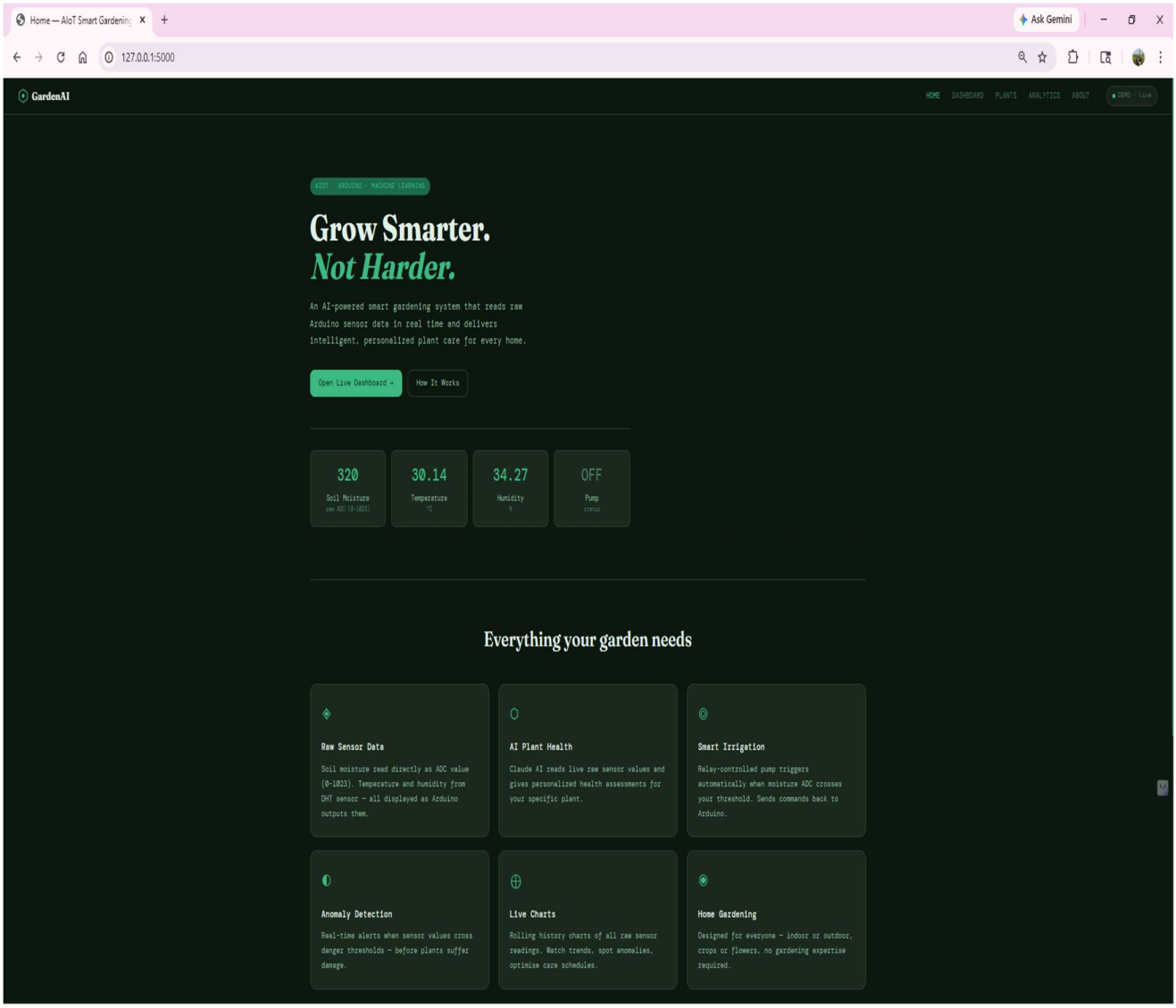
7.2 Live Dashboard

7.3 Plants Profile

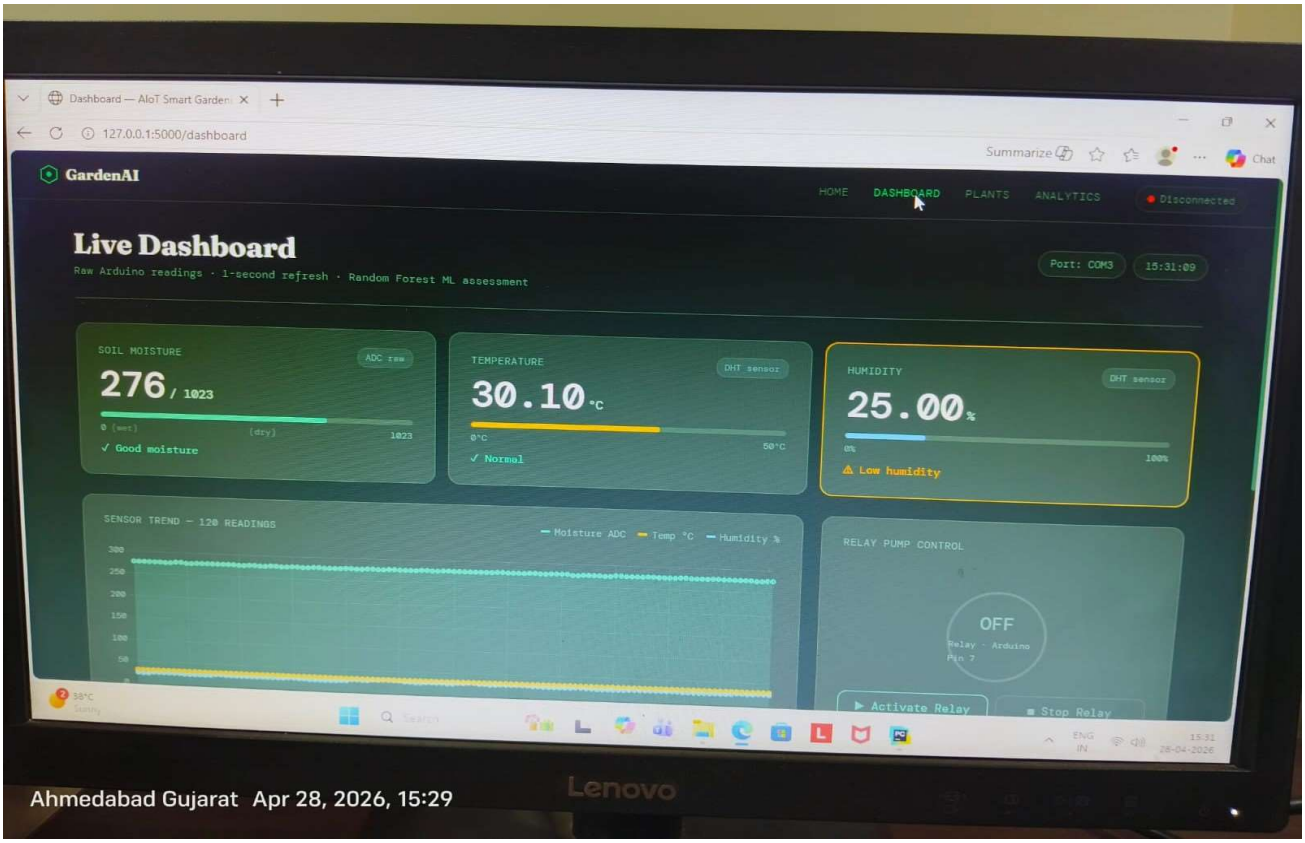
7.4 Analytics

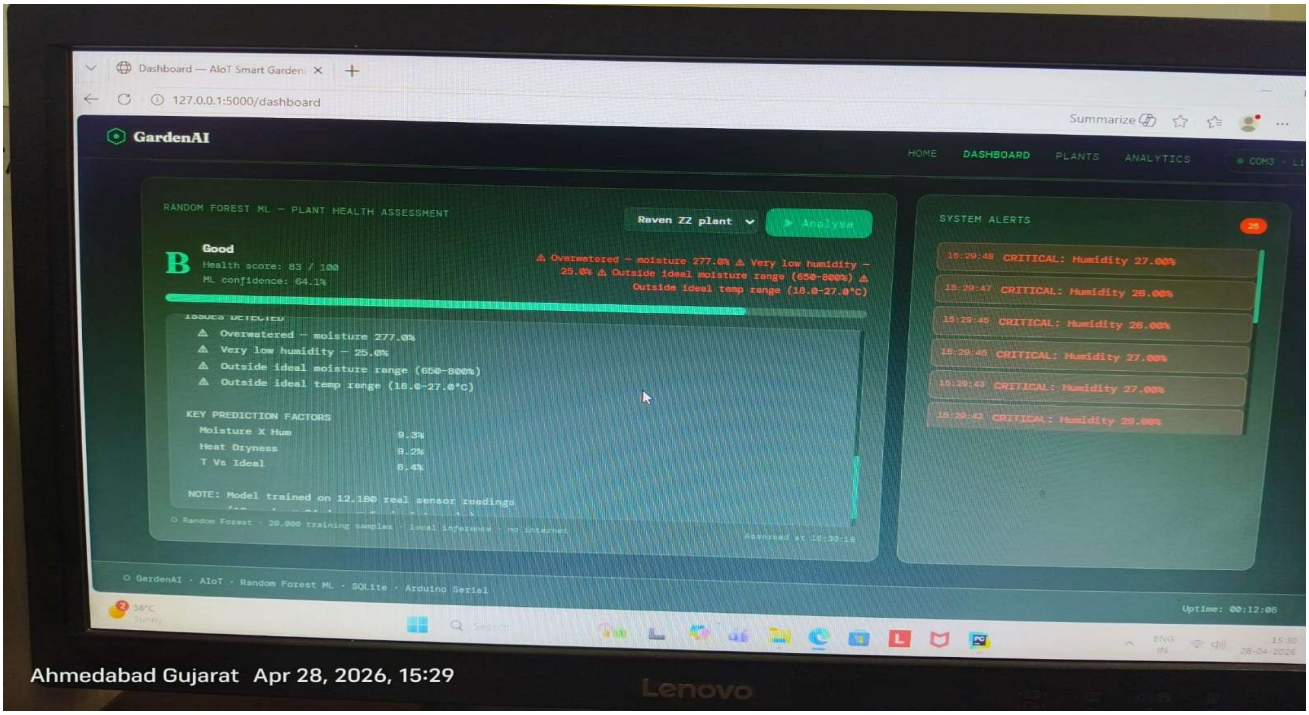
7.5 Arduino system

7.1 Home Page

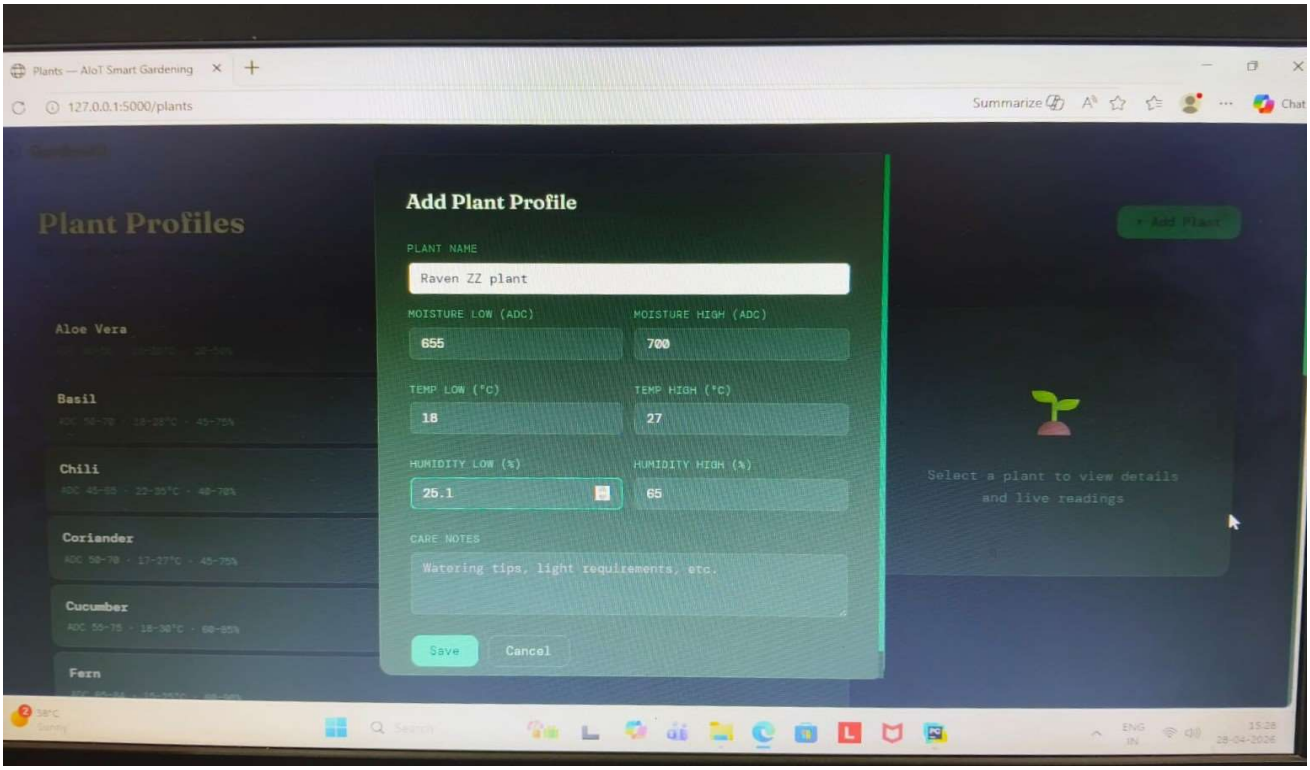

Home — AIoT Smart Gardening
Ask Gemini
127.0.0.1:5000
GardenAI
HOME
DASHBOARD
PLANTS
ANALYTICS
ABOUT
AIOT · ARDUINO · MACHINE LEARNING
Grow Smarter.
Not Harder.
An AI-powered smart gardening system that reads raw Arduino sensor data in real time and delivers intelligent, personalized plant care for every home.
Open Live Dashboard →
How It Works
320
Soil Moisture
30.14
Temperature
34.27
Humidity
OFF
Pump
Everything your garden needs
Raw Sensor Data
Soil moisture read directly as ADC value (0–1023). Temperature and humidity from DHT sensor — all displayed as Arduino outputs them.
AI Plant Health
Claude AI reads live raw sensor values and gives personalized health assessments for your specific plant.
Smart Irrigation
Relay-controlled pump triggers automatically when moisture ADC crosses your threshold. Sends commands back to Arduino.
Anomaly Detection
Real-time alerts when sensor values cross danger thresholds — before plants suffer damage.
Live Charts
Rolling history charts of all raw sensor readings. Watch trends, spot anomalies, optimise care schedules.
Home Gardening
Designed for everyone — indoor or outdoor, crops or flowers, no gardening expertise required.

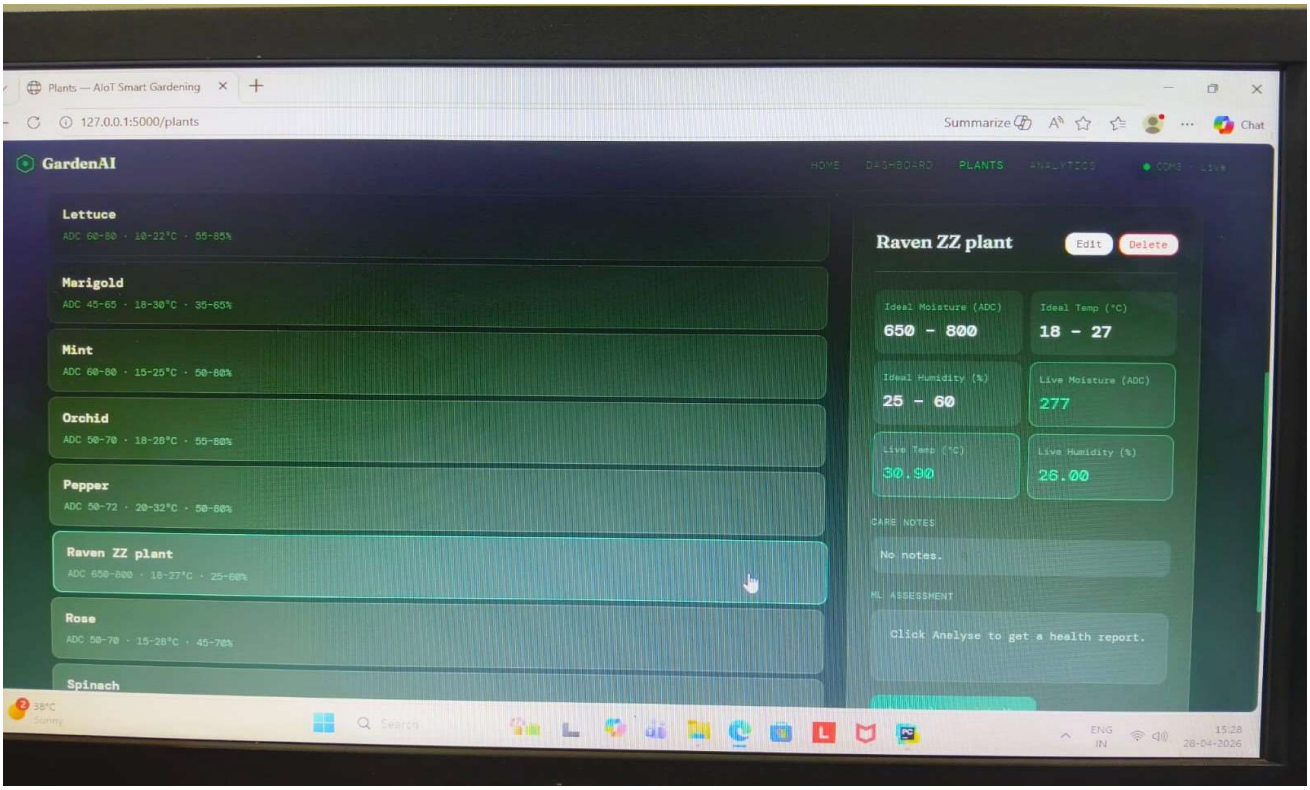
7.2 Live Dashboard:



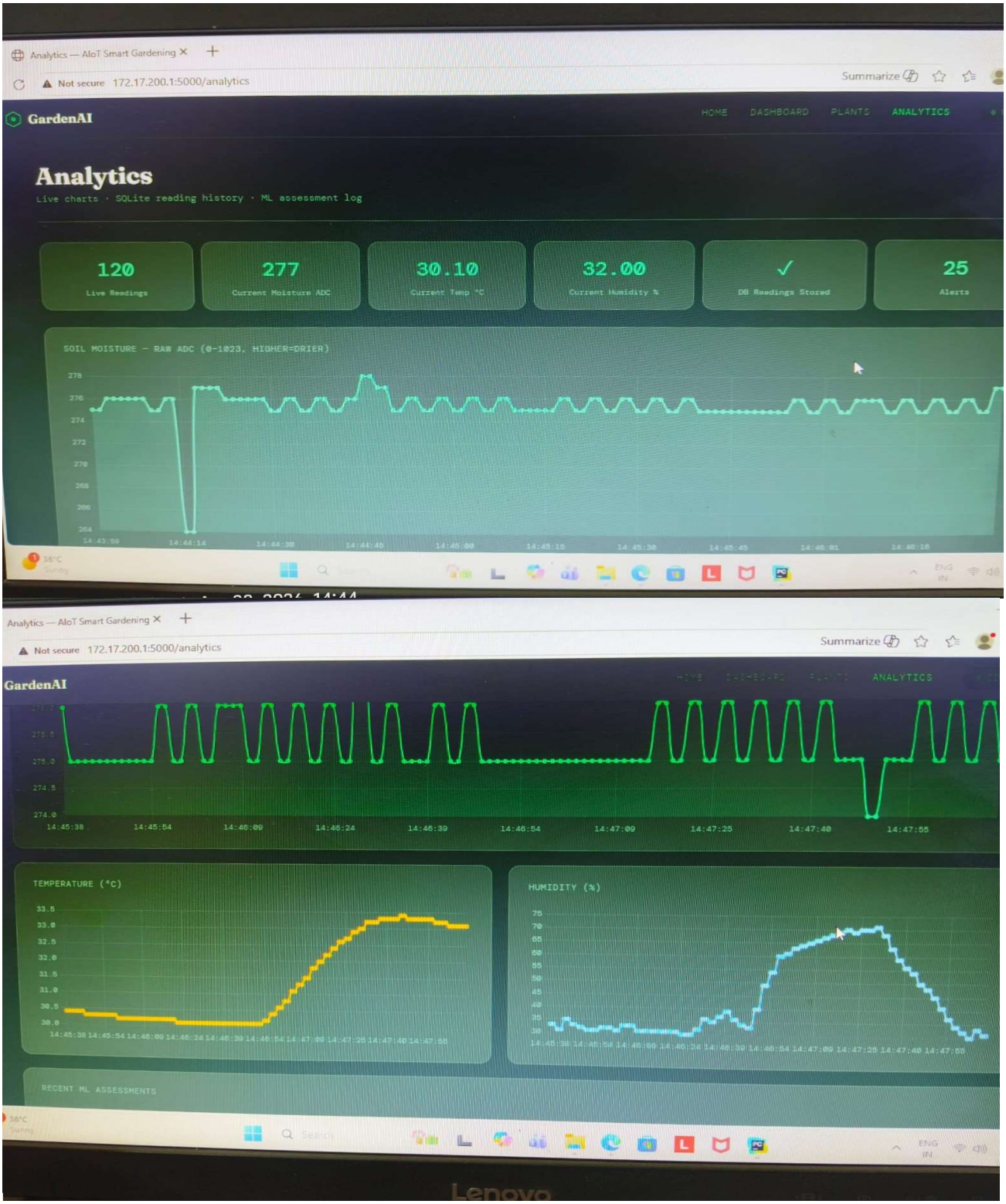


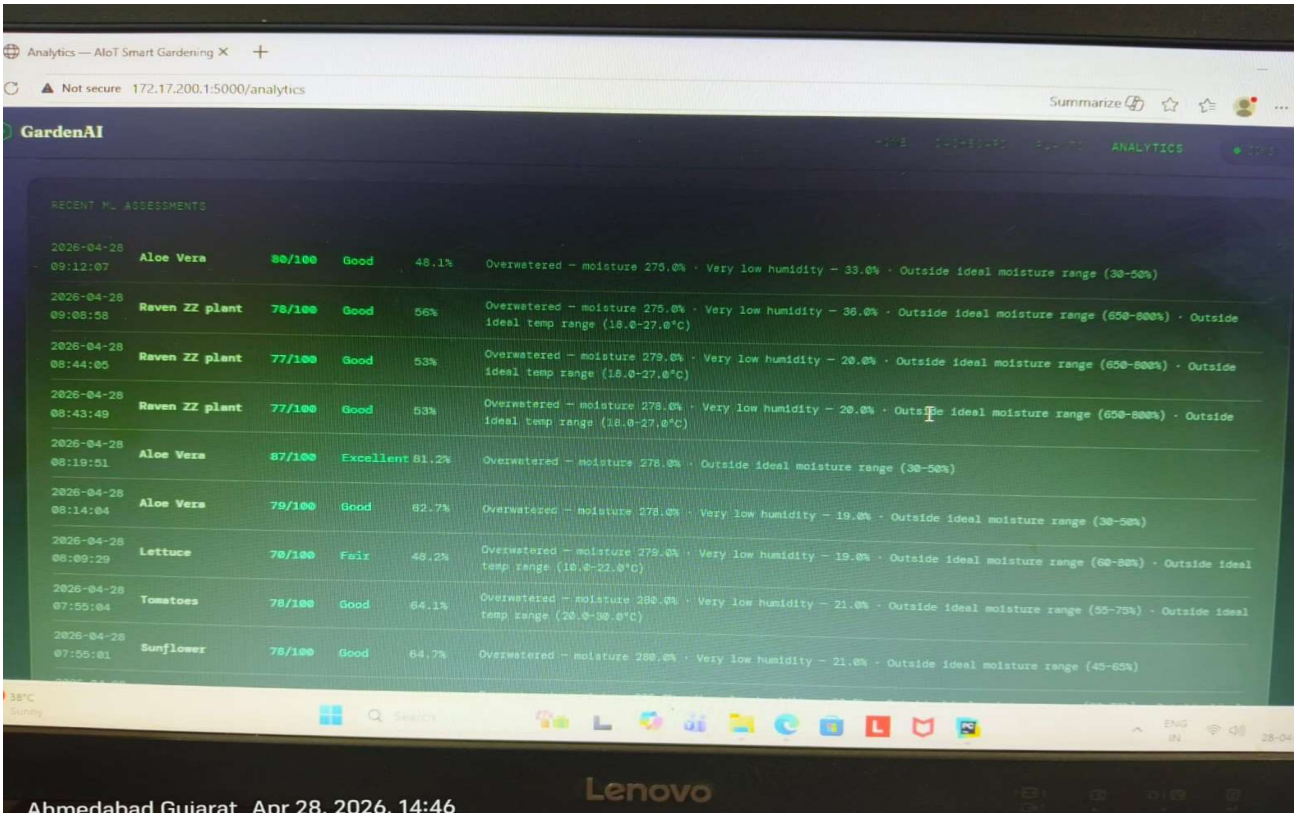
7.3 Plants Profile



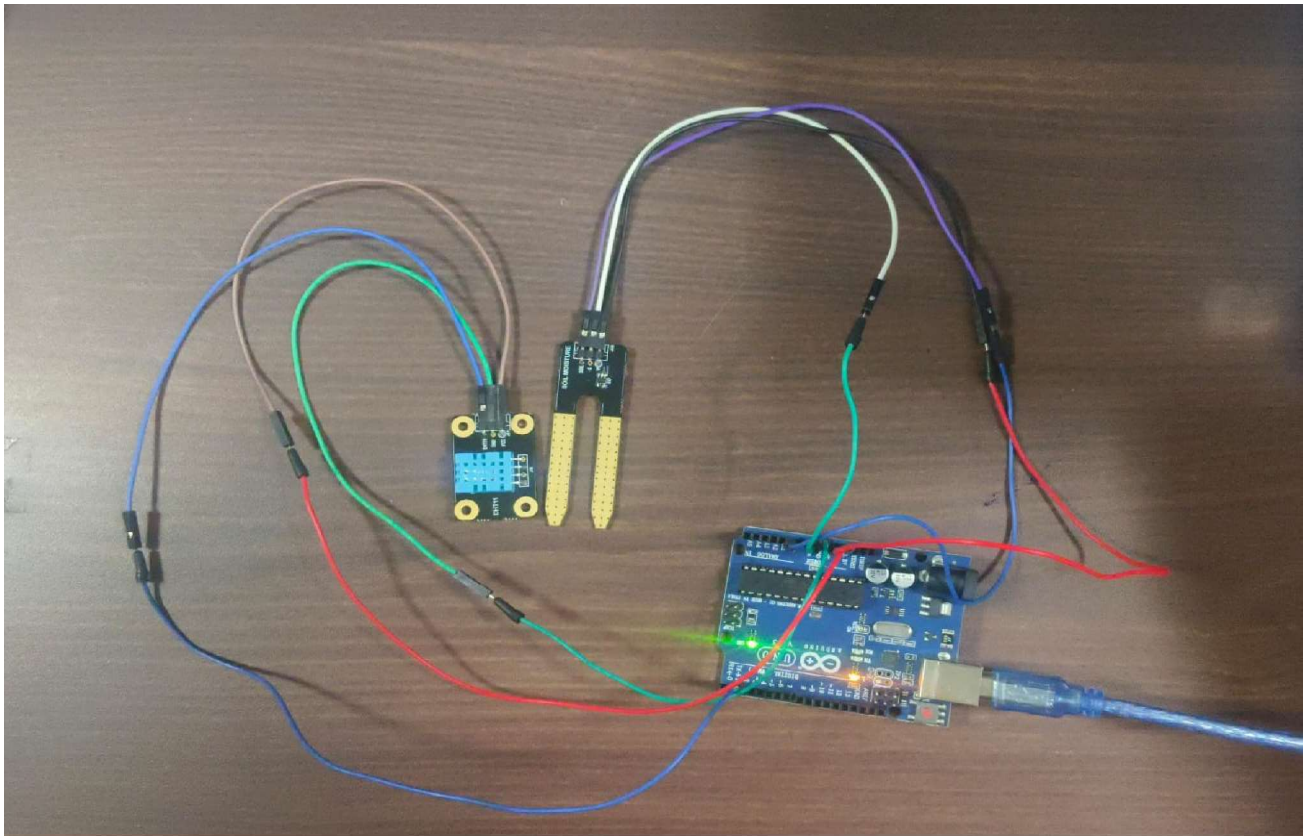


7.4 Analytics




Analytics — AIoT Smart Gardening
Not secure 172.17.200.1:5000/analytics
Summarize
GardenAI
ANALYTICS
RECENT ML ASSESSMENTS
2026-04-28 09:12:07 Aloe Vera 80/100 Good 48.1% Overwatered — moisture 275.0% · Very low humidity — 33.0% · Outside ideal moisture range (30–50%)
2026-04-28 09:08:58 Raven ZZ plant 78/100 Good 56% Overwatered — moisture 275.0% · Very low humidity — 36.0% · Outside ideal moisture range (650–800%) · Outside ideal temp range (18.0–27.0°C)
2026-04-28 08:44:05 Raven ZZ plant 77/100 Good 53% Overwatered — moisture 279.0% · Very low humidity — 20.0% · Outside ideal moisture range (650–800%) · Outside ideal temp range (18.0–27.0°C)
2026-04-28 08:43:49 Raven ZZ plant 77/100 Good 53% Overwatered — moisture 278.0% · Very low humidity — 20.0% · Outside ideal moisture range (650–800%) · Outside ideal temp range (18.0–27.0°C)
2026-04-28 08:19:51 Aloe Vera 87/100 Excellent 81.2% Overwatered — moisture 278.0% · Outside ideal moisture range (30–50%)
2026-04-28 08:14:04 Aloe Vera 79/100 Good 62.7% Overwatered — moisture 278.0% · Very low humidity — 19.0% · Outside ideal moisture range (30–50%)
2026-04-28 08:09:29 Lettuce 70/100 Fair 48.2% Overwatered — moisture 279.0% · Very low humidity — 19.0% · Outside ideal moisture range (60–80%) · Outside ideal temp range (10.0–22.0°C)
2026-04-28 07:55:04 Tomatoes 78/100 Good 64.1% Overwatered — moisture 280.0% · Very low humidity — 21.0% · Outside ideal moisture range (55–75%) · Outside ideal temp range (20.0–30.0°C)
2026-04-28 07:55:01 Sunflower 78/100 Good 64.7% Overwatered — moisture 280.0% · Very low humidity — 21.0% · Outside ideal moisture range (45–65%)
Lenovo
Ahmedabad Gujarat, Apr 28, 2026, 14:46

7.5 Arduino System



CHAPTER: 8

ADVANTAGES & LIMITATIONS

8.1 Advantages

8.2 Limitations

8.1 Advantages

- **Automated Irrigation:** The system automatically controls watering based on soil moisture levels, eliminating the need for manual intervention. This ensures plants receive water only when required.
- **Water Conservation:** By using real-time soil data, the system prevents overwatering and water wastage, making it environmentally sustainable and efficient.
- **Real-Time Monitoring:** Sensors continuously monitor: Temperature, Humidity, Soil Moisture. This allows users to track plant conditions through the dashboard at any time.
- **Improved Plant Health:** With proper monitoring and timely irrigation:
    1) Plants grow healthier.
    2) Risk of under/over watering is reduced.
    3) Environmental conditions remain optimal.
- **Remote Access & Control:** Using a dashboard (Streamlit/web app), users can Monitor data remotely and Control irrigation system manually.
- **Scalability:** The system can be extended by adding:
    1) More sensors.
    2) Multiple irrigation zones.
    3) Cloud integration (IoT platforms).
- **Data Collection for Analysis:** Stores environmental data which can be used for:
    1) Trend analysis.
    2) Predictive modeling.
    3) Improving plant care strategies.

8.2 Limitations

- **Efficiency:** The present study demonstrates the effectiveness of a Random Forest and sensor approach for Smart Gardening System. However, several limitations should be acknowledged.
- **Dependency on Power Supply:** The system requires continuous power. The Power failure can stop irrigation and also it may affect monitoring system.
- **Limited Range of Arduino:** Arduino Uno:

    1) Has limited memory and processing power.
    2) Not suitable for large-scale or complex AI processing.

- **Environmental Sensitivity:** Sensors may be affected by:

    1) Extreme weather conditions.
    2) Dust, corrosion, or water exposure.

- **Maintenance Requirement:** Regular maintenance is needed:

    1) Cleaning sensors.
    2) Checking connections.
    3) Replacing faulty component.s

- **No Native Internet Connectivity:** Arduino Uno does not have built-in WiFi:

    1) Requires additional modules (ESP8266/ESP32).
    2) Limits real IoT functionality.

- **Initial Setup Complexity:** Although components are simple:

    1) Integration of hardware + software + AI can be complex.
    2) Requires knowledge of multiple domains.

- **Limited Coverage Area:** Suitable mainly for home gardening or small farms, it is not ideal for large agricultural fields without scaling.

CHAPTER: 9

CONCLUSION AND FUTURE ENHANCEMENT

9.1 Conclusion

9.2 Future Enhancement

9.1 Conclusion

The Smart Gardening System developed in this project successfully demonstrates the integration of IoT (Internet of Things) and Artificial Intelligence (AI) for efficient and automated plant care. By utilizing sensors such as soil moisture and DHT11 (temperature and humidity), the system continuously monitors environmental conditions and makes intelligent decisions regarding smart irrigation.

The use of an Arduino Uno as the central controller enables real-time data acquisition and control of the irrigation system through a relay module. This ensures that plants receive adequate water based on actual soil conditions, thereby reducing water wastage and human effort works efficiently.

Furthermore, the incorporation of AI-based disease detection and plant health prediction enhances the system's capability by providing early insights into plant conditions. The dashboard interface allows users to monitor and control the system easily, making it user-friendly and practical for home gardening applications.

Overall, the system achieves:

- Efficient water management
- Improved plant health monitoring
- Reduced Human intervention
- Integration of hardware and intelligent software

This project proves to be a cost-effective and scalable solution for modern smart agriculture and sustainable gardening practices.

9.2 Future Enhancement

Although the current system is functional and efficient, several enhancements can be implemented to improve its performance and scalability:

- **Integration with IoT Cloud Platforms:**
  1. Use platforms like ThingSpeak, Blynk, or AWS IoT
  2. Enable remote monitoring from anywhere in the world
  3. Store large-scale data for analysis

- **Use of ESP32 / NodeMCU:**
  1. Replace Arduino Uno with ESP32
  2. Built-in WiFi and Bluetooth
  3. Better processing power and connectivity

- **Advanced AI Models**
  1. Improve disease detection using deep learning (CNN models)
  2. Train models with larger datasets
  3. Increase prediction accuracy

- **Mobile Application Development**
  1. Develop an Android/iOS app
  2. Provide notifications and alerts

- **Weather Forecast Integration**
  1. Use APIs to fetch weather data
  2. Adjust irrigation schedule based on rainfall prediction

- **Multi-Zone Irrigation System**
  1. Control multiple plant areas independently
  2. Useful for larger gardens or farms
- **Fertilizer Automation (Smart Nutrient System)**
  1. Add nutrient sensors
  2. Automate fertilizer supply along with irrigation

- **Enhanced Data Analytics Dashboard:** Adding new features -
  1. Predictive analytics
  2. Growth trends
  3. Smart recommendations

BIBLIOGRAPHY

a. **Course Outcome**

b. **Books**

c. **Web Reference**

a. Course Outcome

Sr. No.	CO statement	Marks % weightage	Chapter No.
CO-1	Undertake problem identification, formulation and solution	20%	**CHAPTER 1** INTRODUCTION
CO-2	Design engineering solutions to complex problems utilising a systematic approach and team work	30%	**CHAPTER 2** SYSTEM REQUIREMENTS **CHAPTER 3** DAILY TASK
CO-3	Communicate with engineers and the community at large in written and oral forms	20%	**CHAPTER 8** ADVANTAGES **CHAPTER 9** CONCLUSION AND FUTURE ENHANCEMENT
CO-4	Demonstrate the knowledge and understanding of engineering and management principle and apply it to assigned project	30%	**CHAPTER 4** AI MACHINE LEARNING MODEL **CHAPTER 5** HARDWARE COMPONENTS **CHAPTER 6** DATA DICTIONARY

b. Books:

- **Intelligent and automatic irrigation system based on internet of things using fuzzy control technology** :
  - Authors: Xinying Liu, Zhihuan Zhao, Amin Rezaeipanah-2025.
- **Smart irrigation system using PLC, SCADA, and fuzzy control:**
  - Author: Iman Morsi, Maged Ahmed-2024
- **Smart Drip Irrigation System Based on IOT Using Fuzzy Logic**
  - Author: Lynn Beighley & Michael Morrison
- **Fuzzy Logic Design to Control the Duration of Irrigation Time in the Greenhouse**
  - Author: D. D. K. Widyawati, A. A. Ambarwari-2022

c. Web Reference:

- **The datasets used during the current study:**
  - Available: figshare.com/s/aab5b90fd151f5ecd8e1.
- **Arduino codes and system installation help:**
  - Available: Arduino Docs | Arduino Documentation.
- **Finding research paper related to project**
  - [Online] Available: IEEE Xplore
- **Stack Overflow. "Where Developers Learn, Share, & Build Careers." [Online].**
  - Available: https://stackoverflow.com/